

Constructor

Definition

A **constructor** in Java is a special method that is used to initialize objects.

- It has the same name as the class.
 - It does not have any return type (not even void).
 - It is automatically called when an object is created.
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Types of Constructors

1. **Default Constructor** → No arguments, assigns default values.
 2. **Parameterized Constructor** → Takes parameters to assign values.
 3. **Constructor Overloading** → Multiple constructors in one class with different parameter lists.
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Example

```
class Student {  
    int id;  
    String name;  
    // Default Constructor  
    Student() {  
        id = 0;  
        name = "Unknown";  
    }  
    // Parameterized Constructor  
    Student(int i, String n) {  
        id = i;  
        name = n;  
    }  
    void display() {  
        System.out.println(id + " " + name);  
    }  
}
```

```
class Main {  
    public static void main(String[] args) {  
        Student s1 = new Student();  
        Student s2 = new Student(101, "Shrushti");  
        s1.display();  
        s2.display();  
    }  
}
```

Key Points for 5 Marks

- Constructor = Special method to initialize objects.
- Same name as class, no return type.
- Called automatically when object created.
- Can be **default / parameterized / overloaded**.
- Helps in **object initialization** easily.